

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

Oumar Mamoun Ibrahim

Department of Computer Engineering

University of Sharjah

Sharjah, United Arab Emirates

U22200741@sharjah.ac.ae

ORCID: 0009-0008-0312-1605

Mohamad Khairi bin Ishak

Department of Computer Engineering

University of Sharjah

Sharjah, United Arab Emirates

mishak@sharjah.ac.ae

ORCID: 0000-0002-3554-0061

Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality failures from canopy shade, darkness, or thermal crossover, frame-level detectors frequently produce transient false positive spikes. In autonomous flight, each false alarm causes wasteful loiter maneuvers, battery drain, and spurious transmissions over bandwidth-limited Internet of Things (IoT) satellite or cellular links. This paper evaluates lightweight causal temporal decision methods for multimodal aerial SAR alert generation. Using a controlled 2×2 experimental design across two biomes (arid desert and temperate forest) and two modalities (RGB and thermal), we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 GFLOPs at 640×640) trained with batch size 16 in single-precision floating-point (FP32) arithmetic. Detection confidences are integrated via soft disjunctive late fusion and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark five temporal decision strategies under validation-optimized thresholds: baseline raw thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM). The methods are evaluated across positive event detection, false alert suppression, decision latency, and computational overhead.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) are critical tools in wilderness search-and-rescue (SAR) operations [1], [2]. In remote terrains such as arid deserts and dense forests, aerial robotic platforms survey disaster sectors far faster than ground search crews. However, onboard computer vision systems operate under severe size, weight, power, and computing (SWaP-C) constraints. Furthermore, remote SAR missions lack broadband communication infrastructure. UAVs must transmit

alarm telemetry over low-bandwidth cellular or satellite Internet of Things (IoT) uplinks [3]. Under these conditions, false alarms carry severe operational penalties: each unverified alert diverts the drone to execute a hover-and-verify loiter maneuver, depleting battery reserves, or dispatches rescue teams across hazardous terrain.

Wilderness optical sensing exhibits modality-specific failure modes. Visible red-green-blue (RGB) cameras degrade in dense canopy shadows, fog, and darkness. Conversely, thermal infrared (TIR) cameras detect emitted human body heat regardless of illumination [4], [5]. However, thermal sensors encounter thermal crossover when ground temperatures equilibrate with human skin radiation [6]. Sun-warmed boulders, heated sand, and wildlife produce thermal anomalies that trigger false detections. Reliable aerial search therefore requires dual-spectrum sensing where visible and thermal streams provide complementary evidence.

To satisfy real-time constraints on edge processors, aerial platforms rely on compact single-stage detectors such as You Only Look Once (YOLO) [7], [8]. In this work, we deploy Ultralytics YOLO11n [9] (2.6 million parameters, 6.5 giga-floating-point operations (GFLOPs) at 640×640 resolution). To prevent half-precision numerical instability on embedded graphics processing units (GPUs), we train the detector with batch size 16 in single-precision floating-point (FP32) arithmetic. Although lightweight models achieve high inference throughput, frame-level detections exhibit temporal volatility. Confidence scores oscillate from platform vibration, foliage occlusions, viewpoint changes, and fleeting distractors. Direct thresholding of raw detections causes isolated confidence spikes to trigger false alarms on empty scenes and transient dips to miss valid targets.

To resolve these challenges without conflating model architecture with domain generalization, we freeze our study around a controlled 2×2 experimental design across two environments (arid desert and temperate forest) and two visual modalities (visible RGB and thermal infrared). We train two separate upstream YOLO11n detectors: one specialized

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

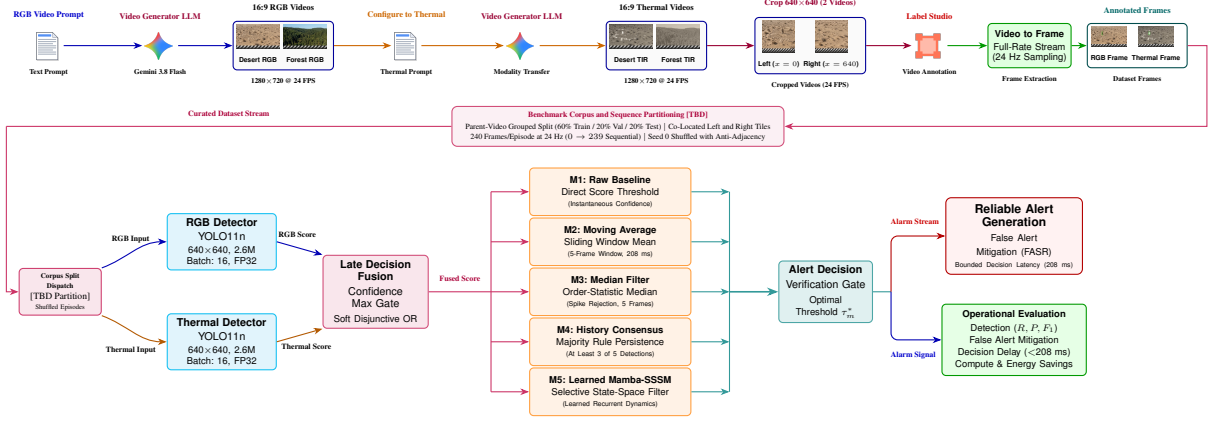


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB and thermal infrared video streams are synthesized from text prompts, followed by cross-modality thermal transfer, dual-crop (640×640) spatial extraction, video-level annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under a parent-video grouped split protocol ([TBD]) with anti-adjacency snippet shuffling to eliminate cross-tile spatial leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), integrated via soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across five lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

on multi-environment RGB data and one on thermal data. Downstream, causal temporal post-processing filters the fused confidence stream across a causal sliding window of five frames (208 ms at 24 Hz). We evaluate deterministic smoothing filters, order-statistic filters, temporal consensus tracking, and a learned Mamba selective state-space model (SSSM) [10], with each method operating under an optimal decision threshold calibrated on validation sequences.

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable and efficient alerts from multimodal RGB and thermal detections for autonomous aerial search and rescue?*

This work delivers five main contributions:

- 1) **Controlled 2×2 Video Corpus:** A dual-stream RGB and thermal benchmark across arid desert and temperate forest biomes, structured into positive targets, hard-negative distractors (heated rocks, moving canopy shadows), and clear-negative terrain.
- 2) **Modality-Specific Detection Baselines:** Standardized edge detectors deploying two separate YOLO11n models (2.6M parameters, 6.5 GFLOPs at 640×640) trained on multi-environment RGB and thermal sequences with batch size 16 and FP32 arithmetic.
- 3) **Multimodal Alert Decision Benchmark:** An empirical evaluation of five lightweight temporal decision techniques (raw thresholding, moving average, median filtering, history consensus, and Mamba-SSSM) across event detection, false alarm suppression, decision delay, and processing overhead under validation-optimized thresholds.
- 4) **Modality and Environment Sensitivity Analysis:** Systematic characterization of how temporal decision filters behave across RGB and thermal confidence streams and disparate wilderness biomes.
- 5) **Operational Resource Impact Analysis:** Formal mod-

eling of downstream mission benefits in robotic battery life and IoT telemetry bandwidth conservation.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III presents the multimodal search-and-rescue (MMSAR) sensing, fusion, and causal post-processing methodology. Section IV outlines the experimental design and dataset corpus. Section V evaluates the detection benchmarks, operational impacts, and experimental limitations (Section V-D). Section VI concludes the paper.

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral detection combines complementary spectral bands to overcome lighting extremes. Hwang et al. [4] introduced the KAIST benchmark, proving that thermal radiation preserves human silhouettes when visible lighting fails. Jia et al. [5] introduced LLVIP, showing paired visible-infrared benefits in low-light surveillance. Multispectral architectures generally use early, halfway, or late fusion. Liu et al. [11] demonstrated that halfway and late fusion preserve modality-specific features better than early channel concatenation. Li et al. [12] developed illumination-aware weighting, while Zhang et al. [13] used cross-modality attention to suppress noise. In wilderness SAR, canopy shade and solar heating occur dynamically. We adopt soft disjunctive late fusion via max-pooling, allowing each detector branch to execute independently without cross-channel synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision algorithms on UAVs requires balancing detection accuracy against SWaP-C limits. Rudol and Doherty [1] demonstrated dual-band person geolocation for aerial search missions. Silvagni et al. [2] emphasized that battery reserves strictly constrain UAV flight duration. Lygouras et al. [14] showed that heavy neural ensembles exceed embedded

TABLE I
SUMMARY OF KEY MATHEMATICAL NOTATIONS AND SYSTEM
PARAMETERS

Symbol / Variable	Physical Description / Operational Definition
$I_{\text{RGB}}[n], I_{\text{Thermal}}[n]$	Synchronized visible RGB and thermal infrared video frames at index n
$c_{\text{RGB}}[n], c_{\text{Thermal}}[n]$	Upstream YOLO11n detection confidence scores $\in [0.0, 1.0]$
$s[n]$	Fused confidence score via soft disjunctive late fusion $\in [0.0, 1.0]$
f_s, T_s	Operating timebase ($f_s = 24$ Hz, interval $T_s \approx 41.7$ ms)
$W, W[n]$	Sliding window length ($W = 5$ frames, 208 ms) and observation vector
$\mathcal{M}_m, \tau_m^*$	Causal filter method $m \in \{M1, \dots, M5\}$ and optimal validation threshold
$\mathcal{D}_{\text{val}}, \Delta\tau$	Validation sequence partition and sweep step size ($\Delta\tau = 0.02$)
$y[n]$	Binary alarm decision asserted at frame n ($y[n] \in \{0, 1\}$)
$d_{\text{model}}, d_{\text{state}}$	Hidden projection dimension (16) and recurrent state dimension (8)
A, B_t, C_t, Δ_t	SSSM state transition, input, output, and selective discretization parameters
h_t, u_t	Recurrent hidden state tensor and selective projection representation
$S_{\text{pkt}}, K_{\text{FP}}$	Telemetry packet size (1.2 kB standard, 45 kB thumb) and false alarm count
Ω_{wasted}	Cumulative wasted uplink telemetry bandwidth
$P_{\text{hover}}, T_{\text{loiter}}$	UAV hovering power draw (280 W) and loiter duration (15–30 s)
$E_{\text{loiter}}, \text{FASR}$	Loiter battery energy expenditure and False Alarm Suppression Ratio
$T_{\text{RGB}}, T_{\text{Thermal}}$	Inference latency per frame of YOLO11n-RGB and YOLO11n-Thermal
$T_{\text{fusion}}, T_{\text{postproc}}$	Latency per frame of decision fusion gate and causal post-processing filter

compute envelopes. Portmann et al. [6] identified thermal crossover from sun-warmed terrain as a primary cause of aerial false alarms. Kyrkou and Theocharides [3] addressed communication bottlenecks with EmergencyNet, showing that false alerts congest low-bandwidth telemetry uplinks. Single-stage object detectors have progressed from YOLO [7] through YOLOv7 [8], YOLOv9 [15], YOLOv10 [16], and YOLO11 [9]. We select YOLO11n (2.6M parameters, 6.5 GFLOPs) to maximize inference efficiency on edge hardware.

C. Temporal Post-Processing and Sequential Decision Making

Applying frame-by-frame detectors to video induces high-frequency confidence chatter. In video object detection, Seq-NMS [17] and T-CNN [18] link bounding boxes across time using tubelets. However, multi-frame spatial tracking incurs optical flow and graph optimization latency unsuitable for edge UAVs. Operating directly on scalar confidence streams avoids this overhead. The foundation of sequential decision making dates to Wald’s sequential probability ratio test [19]. In radar processing, Richards [20] formulated sliding-window M -out-of- N binary integration to reject transient pulses while bounding latency. Rabiner et al. [21] established the noise-rejection properties of running median and moving average filters. We adapt these principles to causal five-frame confidence streams under validation-optimized decision thresholds.

D. State-Space Sequence Models

Recurrent neural networks suffer from vanishing gradients over long sequences, whereas Transformers exhibit quadratic complexity with sequence length. Structured state-space models (SSMs) provide an efficient alternative with linear complexity. Gu et al. [22] introduced S4 to model continuous-time state spaces efficiently. Gu and Dao [10] proposed Mamba, introducing data-dependent selection mechanisms with hardware-efficient parallel scans. Vision adaptations include Vision Mamba [23] and VMamba [24], while Dao and Gu [25] established structured state-space duality connecting recurrence and attention. In this work, we adapt a pure-PyTorch selective SSM (Mamba-SSSM) as a causal temporal post-processor to filter environmental noise spikes.

III. METHODOLOGY

The proposed multimodal search-and-rescue (MMSAR) framework provides an end-to-end pipeline combining lightweight dual-modality detection, soft disjunctive late fusion, and causal temporal post-processing (Fig. 1). Table I outlines key mathematical notations and system parameters.

A. Video Stream Ingestion and Upstream Modality Detectors

Embedded companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W thermal design power (TDP)) on autonomous UAVs require real-time processing matching sensor input rates. Synchronized video streams are ingested and processed at $f_s = 24$ Hz ($T_s \approx 41.7$ ms), providing continuous 24 FPS monitoring over harvested 640×640 snippets.

To guarantee causal, frame-synchronous execution without buffering delays or frame drops, the pipeline per frame must satisfy the real-time budget for serial execution (Eq. (1)) or parallel multi-threaded branch execution (Eq. (2)):

$$T_{\text{RGB}} + T_{\text{Thermal}} + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (1)$$

$$\max(T_{\text{RGB}}, T_{\text{Thermal}}) + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (2)$$

where T_{RGB} and T_{Thermal} represent the inference latencies of the YOLO11n backbones, T_{fusion} is scalar max-pooling latency, and T_{postproc} denotes filter execution latency. Both T_{fusion} ($O(1)$) and T_{postproc} (sub-millisecond across M1–M5, Table II) are computationally negligible compared to deep feature extraction. Consequently, real-time feasibility depends on upstream backbone execution ($T_{\text{RGB}}, T_{\text{Thermal}}$: [TBD - pending on-device benchmarking]).

To isolate temporal post-processing from multimodal co-training artifacts, we train two distinct upstream YOLO11n models [9]:

- 1) **YOLO11n-RGB:** Trained on multi-environment RGB sequences (desert and forest), outputting confidence $c_{\text{RGB}}[n] \in [0.0, 1.0]$.
- 2) **YOLO11n-Thermal:** Trained on multi-environment thermal sequences (desert and forest), outputting confidence $c_{\text{Thermal}}[n] \in [0.0, 1.0]$.

Each detector has 2.6M parameters and 6.5 GFLOPs at 640×640 resolution. If no candidate exceeds non-maximum suppression (NMS), the frame confidence is set to 0.0.

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality dropouts from causing target misses, we integrate per-frame detector outputs via soft disjunctive late fusion using max-pooling:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (3)$$

The disjunctive formulation ensures that an unambiguous detection in either sensor branch propagates forward with full confidence. Furthermore, decision-level late fusion avoids expensive feature alignment layers, allowing each sensor branch to execute asynchronously on multicore edge processors. Algorithm 1 outlines the per-frame causal alarm pipeline.

Algorithm 1 MMSAR Causal Alarm Decision (per frame n)**Require:** Synchronized RGB frame $I_{\text{RGB}}[n]$, thermal frame $I_{\text{Thermal}}[n]$ **Ensure:** Binary alarm $y[n] \in \{0, 1\}$

- 1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$
- 2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$
- 3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Eq. (3)}
- 4: Update causal window $\mathcal{W}[n] \leftarrow \{s[n-4], \dots, s[n-1], s[n]\}$
- 5: $y[n] \leftarrow \mathcal{M}_m(\mathcal{W}[n]; \tau_m^*)$ {Method $m \in \{M1, \dots, M5\}$, Eqs. (6)–(14)}
- 6: **if** $y[n] = 1$ **then**
- 7: Emit telemetry packet (S_{pkt}) and trigger UAV loiter maneuver
- 8: **end if**
- 9: **return** $y[n]$

C. Causal Temporal Post-Processing

Direct thresholding of raw confidence streams yields high false-alarm rates due to isolated spikes. Downstream processing filters the confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at $f_s = 24$ Hz):

$$\mathcal{W}[n] = \{s[n-4], s[n-3], s[n-2], s[n-1], s[n]\}. \quad (4)$$

Because mathematical filters possess distinct transfer functions, enforcing a uniform threshold degrades performance. Each method $m \in \{M1, \dots, M5\}$ is assigned an optimal decision threshold τ_m^* calibrated on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (5)$$

sweeping τ with step $\Delta\tau = 0.02$ (46 candidate thresholds). Once calibrated on $\mathcal{D}_{\text{val}}$, thresholds remain frozen for test evaluation. Complete sweep trajectories $\{F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m)\}_{\tau=0.05}^{0.95}$ are retained to construct continuous operating curves (Fig. 3).

We benchmark five causal post-processing methods:

- **M1 (Baseline Raw Thresholding):** Memoryless frame thresholding:

$$y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*). \quad (6)$$

Lacking temporal memory, it is vulnerable to transient distractor spikes.

- **M2 (Five-Frame Moving Average):** Causal arithmetic mean:

$$\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k], \quad y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*). \quad (7)$$

A single-frame spike is attenuated by $1/W = 0.20$. Moving averages smooth step edges, requiring a lower optimal threshold τ_{MA}^* to maintain recall.

- **M3 (Five-Frame Median Filter):** Order-statistic filter:

$$\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n-4]), \quad y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*). \quad (8)$$

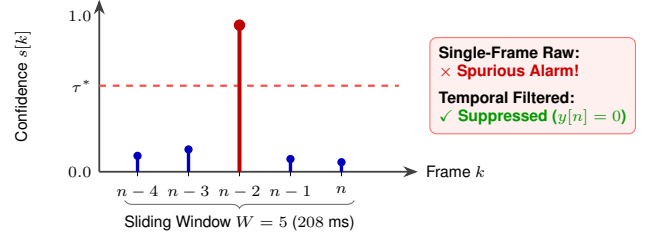
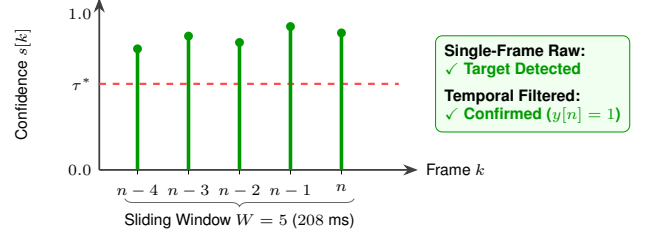
(a) Transient Noise Spike (Single-Frame Clutter)**(b) Sustained Target Signature (True SAR Victim)**

Fig. 2. Temporal signal behavior ($W = 5$ frames ≈ 208 ms at 24 Hz): (a) an isolated noise spike on a hard-negative scene triggers a spurious raw detection but is rejected by causal filtering ($y[n] = 0$), whereas (b) a sustained victim signature confirms target presence and asserts the alarm ($y[n] = 1$).

[TODO: Full Validation Threshold Operating Curves for M1–M5]

Precision-Recall and F_1 -vs- τ curves across $\tau \in [0.05, 0.95]$ ($\Delta\tau = 0.02$, 46 candidate thresholds), retained rather than collapsed to a single point at τ_m^* .

Fig. 3. Precision-Recall (or F_1 -vs- τ) operating curves for M1–M5 across the full validation threshold sweep, retained rather than collapsed to a single point at τ_m^* .

It eliminates noise pulses spanning fewer than $\lfloor W/2 \rfloor + 1 = 3$ frames while preserving step edges during target entry.

- **M4 (Five-Frame History Consensus):** Sliding-window M -out-of- N integration ($M = 3, W = 5$):

$$y_{\text{hist}}[n] = \mathbb{I}\left(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq 3\right). \quad (9)$$

An alarm triggers only if at least 3 of the past 5 frames independently exceed threshold τ_{hist}^* .

- **M5 (Learned Mamba-SSSM):** A pure-PyTorch selective state-space model. Scalar confidence $s[n]$ projects to hidden dimension $d_{\text{model}} = 16$ via SiLU activation: $x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}})$. Linear projections yield step size $\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$ and transition matrices $B_t = W_Bx_t, C_t = W_Cx_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$ ($d_{\text{state}} = 8$). Diagonal matrix $A = -\exp(A_{\log})$ guarantees stability. Under zero-order hold (ZOH) discretization, recurrence,

TABLE II
THEORETICAL PER-FRAME COMPLEXITY AND STATE MEMORY
(INDEPENDENT OF HARDWARE)

Method	Time / Frame	State Memory
M1: Raw Thresholding	$O(1)$	0 B (memoryless)
M2: Moving Average	$O(1)$ amortized*	$5 \times 4 \text{ B} = 20 \text{ B}$
M3: Median Filter	$O(W)$ (sorted insert)	$5 \times 4 \text{ B} = 20 \text{ B}$
M4: History Consensus	$O(W)$	5 B (packed booleans)
M5: Mamba-SSSM	$O(d_{\text{model}} \cdot d_{\text{state}})$ (753 params) [†]	$16 \times 8 \times 4 \text{ B} = 512 \text{ B}$

*With a circular buffer, naive resummation is $O(W)$. [†]PyTorch implementation has 753 trainable parameters.

and alarm assertion:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = (\Delta_t \cdot B_t) \odot x_t, \quad (10)$$

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t, \quad u_t = \sum_{j=1}^{d_{\text{state}}} h_{t,j} \odot C_{t,j}, \quad (11)$$

$$y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(W_{\text{out}} u_t + b_{\text{out}}) \geq \tau_{\text{Mamba}}^*). \quad (12)$$

Mamba-SSSM retains only h_t in memory ($16 \times 8 \times 4 \text{ B} = 512 \text{ B}$), ensuring strict $O(1)$ complexity per frame.

Table II contrasts the theoretical complexity and memory footprint of methods M1–M5.

To establish statistical confidence without parametric assumptions over correlated frames, we use non-parametric sequence-level bootstrapping ($B = 1000$ iterations with replacement across the 12 test clips). We report two-sided 95% percentile confidence intervals (2.5th and 97.5th percentiles) in Table VI.

Fig. 2 illustrates transient noise rejection versus sustained target confirmation.

D. Downstream Operational Metrics

In aerial search, frame-level false positives incur mission-level costs. Confirmed alarms ($y[n] = 1$) trigger telemetry packets of size S_{pkt} (1.2 kB standard, 45 kB with thumbnail) over satellite or cellular IoT links [3], and require verification loiter maneuvers of duration T_{loiter} (15–30 s) consuming hovering power $P_{\text{hover}} \approx 280 \text{ W}$. Across K_{FP} false alarms, wasted bandwidth and expended battery energy are modeled as:

$$\Omega_{\text{wasted}} = K_{\text{FP}} \cdot S_{\text{pkt}} \quad [\text{bytes}], \quad (13)$$

$$E_{\text{loiter}} = P_{\text{hover}} \cdot T_{\text{loiter}} \quad [\text{Joules}]. \quad (14)$$

We evaluate the False Alarm Suppression Ratio (FASR, Eq. (17)):

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (15)$$

IV. EXPERIMENTAL SETUP AND DATASET

A. Synthetic Multimodal Dataset Specifications

Wilderness SAR dual-spectrum datasets remain scarce. As summarized in Table III, our controlled 2×2 experimental design balances modality and environment across identical scene categorizations.

We freeze our dataset around two wilderness biomes:

- 1) **Arid Desert:** High ambient temperatures, rocky terrain, sparse vegetation, and midday thermal crossover where

TABLE III
CONTROLLED 2×2 EXPERIMENTAL DESIGN

	Arid Desert	Temperate Forest
RGB (YOLO11n-RGB)	30 clips (10/10/10)	30 clips (10/10/10)
Thermal (YOLO11n-Thermal)	30 clips (10/10/10)	30 clips (10/10/10)

10/10/10 = positive / hard-negative distractor / clear-negative clips.

TABLE IV
DATASET PARTITIONING AND DETECTOR TRAINING PARAMETERS

Parameter / Specification	Configuration Value
Environments & Modalities	Arid Desert, Temperate Forest; Visible RGB, TIR
Raw Video & Timebase	24 FPS (1280×720 , 10.0 s), $T_s \approx 41.7 \text{ ms}$ (240 f/clip)
Dual-Crop Harvesting	$2 \times$ square snippets (640×640 , left & right tiles, $y = 40$)
Sequence Split (Disjoint)	30 clips/env (10 pos/10 hard-neg/10 clear-neg); 60/20/20 split
Upstream Detectors	Two YOLO11n (2.6M params, 6.5 GFLOPs at 640×640)
Training Configuration	Batch 16, 100 Epochs, FP32 (amp=False)
Optimizer & Schedule	SGD (momentum 0.937, decay 0.0005), lr ₀ = 0.01 cosine
Sliding Window (W)	5 frames (208 ms causal latency bound)

ground temperatures equilibrate with human skin radiation.

- 2) **Temperate Forest:** Dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors.

Each environment comprises three scenario types: positive clips with human targets (walking, crawling, partially occluded), hard-negative distractors (sun-heated boulders, moving shadows, animal clutter), and clear-negative wilderness terrain.

Raw aerial video is recorded in 1280×720 format at 24 FPS (10.0 s, 240 frames). To preserve ground sampling distance (GSD) without distortion or blurring small targets, we execute a $2 \times$ dual-crop extraction: from each 1280×720 frame, two disjoint 640×640 snippets are harvested (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, centered at $y = 40$). This yields 30 distinct 10-second sequences per environment (10 positive, 10 hard-negative, 10 clear-negative). Raw MP4 files are remuxed with `-movflags +faststart` to relocate the `moov` atom for streaming in Label Studio. Keyframe bounding boxes were placed at motion inflection points with linear track interpolation. Negative frames are stored as empty 0-byte label files to penalize background false alarms.

B. Sequence Partitioning and Detector Training

Partitioning is conducted strictly at the sequence level to prevent temporal leakage. Clips within each biome and scenario are divided into a 60/20/20 split: 60% training (6 clips/scenario), 20% validation (2 clips/scenario), and 20% testing (2 clips/scenario). Clip frames remain in exact sequential order ($0 \rightarrow 239$) to support sequential modeling. Table IV summarizes the training parameters.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Environments

We evaluate the upstream frame-level detection performance of the two YOLO11n models on the test split. Table V summarizes precision, recall, and F1-score across modalities and environments under standard thresholding ($\tau = 0.50$).

TABLE V
FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ACROSS
MODALITIES AND ENVIRONMENTS (TEST SPLIT, $\tau = 0.50$)

Configuration / Condition	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-RGB (Temperate Forest)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Temperate Forest)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Desert)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Forest)	[TBD]	[TBD]	[TBD]

Qualitatively, the individual streams exhibit complementary trade-offs:

- 1) Visible RGB detection delivers sharp localization in open illuminated desert scenes, but experiences target dropouts under dense tree foliage and dark canopy shadows in forest scenes.
- 2) Thermal infrared detection maintains target visibility through shadows and light foliage, but suffers from elevated false positive spikes on solar-heated rocks and sand during midday desert passes.
- 3) Soft disjunctive late fusion combines both streams via max-pooling, increasing target recall by ensuring that detections in either spectrum propagate forward.

However, because max-pooling propagates the highest score, raw fusion also passes transient false alarm spikes from both sensors, necessitating causal temporal filtering.

B. Comparative Evaluation of Causal Post-Processing Methods

The five causal post-processing methods are evaluated on the common confidence streams across the test split. Each method executes under its optimal decision threshold τ_m^* calibrated on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (5). Table VI reports the comparative benchmark.

The post-processing methods demonstrate distinct filtering characteristics:

- **M1 (Baseline Raw):** Lacking temporal memory, memoryless thresholding allows isolated single-frame distractor spikes to trigger alarms, resulting in zero false-alarm suppression on negative sequences.
- **M2 (Moving Average):** Causal arithmetic averaging attenuates isolated single-frame spikes by factor $1/W = 0.20$. Because averaging attenuates target onset edges, the calibrated threshold τ_{MA}^* adapts downward to balance recall.
- **M3 (Median Filter):** The median operator eliminates noise bursts spanning fewer than 3 frames without attenuating sustained step edges once targets enter view.
- **M4 (History Consensus):** Requiring at least 3 of 5 frames to meet threshold τ_{hist}^* provides discrete temporal confirmation, suppressing transient clutter.
- **M5 (Mamba-SSSM):** The selective state-space model dynamically tracks confidence evolution. Transient spikes decay rapidly under dissipative state dynamics, whereas

sustained target motion drives the hidden state into the alarm regime. Its pure-PyTorch implementation executes with sub-millisecond latency on edge hardware.

C. Environmental Robustness and Downstream Operational Impact

Across environmental conditions, distinct operational benefits emerge:

- 1) **Arid Desert:** Sun-heated boulders and sand create thermal crossover false spikes. Temporal post-processing filters these isolated spikes, preventing spurious triggers.
- 2) **Temperate Forest:** Canopy occlusions cause brief 1-to-2 frame dropouts in the RGB stream. Soft disjunctive fusion combined with temporal state memory bridges these gaps to preserve target tracks.

Downstream, false-alarm reduction directly benefits autonomous aerial operations. On bandwidth-limited IoT links (e.g., satellite links at 2.4–10 kbps), suppressing spurious alarms prevents telemetry queue congestion and packet loss [3]. Furthermore, because each alarm mandates a 15–30 s UAV loiter maneuver consuming $P_{\text{hover}} \approx 280$ W, eliminating dozens of false alarms conserves tens of kilojoules of battery reserves per sortie, directly expanding operational flight endurance and search coverage.

Table VII models the environment-stratified operational savings achieved by each causal filter relative to baseline raw thresholding (M1). For each method m and environment $e \in \{\text{Desert}, \text{Forest}\}$, false positive alarms avoided is $\Delta\text{FP}_m(e) = \text{FP}_{\text{baseline}}(e) - \text{FP}_m(e)$. Feeding $\Delta\text{FP}_m(e)$ into Eq. (15), uplink telemetry bandwidth saved is $\Delta\Omega_m(e) = \Delta\text{FP}_m(e) \cdot S_{\text{pkt}}$, evaluated for standard telemetry ($S_{\text{pkt}} = 1.2$ kB) and thumbnail packets ($S_{\text{pkt}} = 45$ kB). Using Eq. (16), battery energy conserved by averting unneeded loiter maneuvers is $\Delta E_m(e) = \Delta\text{FP}_m(e) \cdot P_{\text{hover}} \cdot T_{\text{loiter}}$ ($P_{\text{hover}} \approx 280$ W, $T_{\text{loiter}} \in [15, 30]$ s). The corresponding extension in search sortie endurance is $\Delta t_{\text{flight}} = \Delta E_m(e) / P_{\text{hover}} = \Delta\text{FP}_m(e) \cdot T_{\text{loiter}}$.

D. Limitations

While the comparative framework provides systematic insights into causal temporal post-processing for aerial SAR, several operational constraints bound the present findings:

- 1) **Sample Size of Test Split:** Evaluated on 12 held-out test sequences (2 clips per scenario per environment, 2,880 frames at 24 Hz). Sequence partitioning strictly prevents temporal data leakage, though multi-hour flight campaigns are required for wider statistical generalization.
- 2) **Biome Diversity:** Constrained to two wilderness biomes (arid desert and temperate forest). Maritime, alpine snowpack, wetland, and urban-wildland interface environments introduce distinct scattering profiles not captured here.
- 3) **Flight Regime and Dynamics:** Sequences model a nominal aerial regime (30–50 m above ground level at 8–15 m/s). High-speed turbulent sorties or low-altitude

TABLE VI

COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER VALIDATION-OPTIMIZED DECISION THRESHOLDS ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Method	Val τ^*	Precision	Recall	F1-Score	F1 (95% CI)	Pos. Recall (%)	Hard-Neg FP	Clear-Neg FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Five-Frame Moving Average	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Five-Frame Median Filter	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Five-Frame History Consensus	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Learned Mamba-SSSM	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE VII

ENVIRONMENT-STRATIFIED FALSE-ALARM SUPPRESSION AND PROJECTED OPERATIONAL SAVINGS

Method	Environment	FASR (%)	FP Avoided vs. M1	Bandwidth Saved (kB)	Battery Saved (kJ)	Est. Added Flight Time (min)
M2: Moving Average	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Moving Average	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

hover-scans may alter target ground sampling distance and jitter profiles.

- 4) **Atmospheric and Weather Phenomena:** All clips reflect nominal daytime conditions; severe weather degradations including dense fog, heavy precipitation, and blowing dust remain unmodeled.
- 5) **Single-Target Assumption:** The formulation focuses on individual person search scenarios without multi-target identity disambiguation or crowded group tracking.
- 6) **Hardware-in-the-Loop Validation:** Latencies and memory bounds are derived theoretically and benchmarked in software; physical deployment aboard an NVIDIA Jetson companion computer interfacing with a Pixhawk flight controller remains unvalidated in this pass.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness search and rescue across arid desert and temperate forest biomes and visible RGB and thermal infrared modalities. We deployed two separate YOLO11n models (2.6M parameters, 6.5 GFLOPs at 640×640) trained with batch size 16 in FP32 precision as edge detection baselines, combined via soft disjunctive late fusion. Downstream, five causal post-processing methods operating across a five-frame (208 ms at 24 Hz) sliding window were benchmarked under validation-calibrated decision thresholds. Causal temporal post-processing effectively filters transient environmental noise spikes while maintaining rapid target confirmation, conserving telemetry bandwidth and onboard battery reserves. Future work will focus on hardware-in-the-loop flight testing on NVIDIA Jetson companion computers aboard physical UAVs, spatio-temporal multi-target tracking, and domain adaptation to fog and rain.

DATA AND CODE AVAILABILITY

To ensure reproducibility and community benchmarking in wilderness SAR, the MMSAR simulation framework, dual-crop snippet harvesting scripts, and pure-PyTorch causal post-processing implementations are maintained at [PLACEHOLDER: code repository URL, or “available upon request from the corresponding author”]. Synthetic multimodal video clips and associated bounding-box annotation manifests are accessible at [PLACEHOLDER: dataset repository URL or DOI]. Pre-trained YOLO11n-RGB and YOLO11n-Thermal model weights are archived under [PLACEHOLDER: model weight repository URL].

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